Estimate with FIMS : : CHEATSHEET

GET FIMS

```
install.packages("FIMS", repos = c("https://noaa-fisheries-integrated-toolbox.r-universe.dev", "https://cloud.r-project.org"))
```

TIDY THE INPUT FIMSFrame

Prepare one tidy data frame for your FIMS model

```
model_inputs <-
FIMSFrame(your_long_data_frame)
```

Investigate the FIMSFrame class

get_*() functions extract information from a FIMSFrame object

```
data("data_big", package = "FIMS")
fims_inputs <- FIMSFrame(data_big)
get_n_fleets(fims_inputs)
[1] 2
get_ages(fims_inputs)
[1] 1 2 3 4 5 6 7 8 9 10 11 12
get_data(fims_inputs)
```

Package data

Looks like the table below

Structure of the long, data format

type	fleet	timing	value	unit	uncertainty	age
landings	trawl	2025	1000	mt	0	NA
landings	pot	2024	2366	mt	0.05	NA
age_comp	trawl	2025	5	number	100	1
age_comp	trawl	2025	2	number	100	2
...	...	...	...	...	...	

type: landings, index, age_comp, length_comp, weight_at_age, age_to_length_conversion
fleet: name of the fleet, can be any string
timing: double values specifying the collection timing
value: the measurement
unit: unit of the measurement
uncertainty: measure of uncertainty, e.g., standard deviation, sample size
age: desired binning structures for your model, can have additional binning columns such as **length** and, in the future, **sex**

FIT A MODEL FIMSFit

Default parameters based on data

See vignette for details on modifying the parameters object, which is a nested tibble

```
# Create parameters
parameters <- model_inputs |>
  create_default_configurations() |>
  create_default_parameters(data = model_inputs)
```

Some default model configurations are created based on the data, e.g., logistic selectivity for each fleet in your data

Use the configurations to determine the suite of required parameters and set them to their default values, e.g., logistic selectivity requires two parameters, inflection_point and slope (see below)

```
# View parameters
tidyr::unnest(parameters, cols = data) |>
  dplyr::select(-we_removed_several_columns)
# A tibble: 655 × 9
   model_family module_name fleet  module_type label             time value estimation_type
   <chr>        <chr>       <chr>  <chr>       <chr>            <int> <dbl> <chr>
 1 catch_at_age Selectivity fleet1 Logistic    inflection_point    NA     2 fixed_effects
 2 catch_at_age Selectivity fleet1 Logistic    slope               NA     1 fixed_effects
 3 catch_at_age Fleet       fleet1 NA          log_q               NA     0 constant
 4 catch_at_age Fleet       fleet1 NA          log_Fmort            1    -3 fixed_effects
 5 catch_at_age Fleet       fleet1 NA          log_Fmort            2    -3 fixed_effects
# ℹ 650 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Fit a FIMS model

Send all of the specified parameters and data to their respective modules

```
# Initialize the C++ modules and
# fit the model to the data
fit <- parameters |>
  initialize_fims(data = model_inputs) |>
  fit_fims(optimize = TRUE)
```

Fit the model

Investigate the FIMSFit class

Investigate the FIMSFit class get_*() functions extracts information from a FIMSFit object

```
get_estimates(fit)
get_fits(fit)
get_number_of_parameters(fit)
get_timing(fit)
```

```
# TMB information
get_report(fit)
get_obj(fit)
get_par(fit)
```

WARNING: see Discussion #974 about changing these to extract_*()